2.71 3D Matrices: Factorisation of Determinants

Worked Solutions

1 a)

$$|\mathbf{A}| = \begin{vmatrix} 4 & 2 & 6 \\ 0 & 6 & 15 \\ 0 & 0 & 12 \end{vmatrix} = 4(6 \cdot 12 - 15 \cdot 0) - 2(0 \cdot 12 - 15 \cdot 0) + 6(0 \cdot 0 - 6 \cdot 0) = 4 \cdot 6 \cdot 12 = 288.$$

1 b)

$$2\begin{vmatrix} 2 & 1 & 3 \\ 0 & 6 & 15 \\ 0 & 0 & 12 \end{vmatrix} = 2\big(2(6 \cdot 12 - 15 \cdot 0) - 1(0 \cdot 12 - 15 \cdot 0) + 3(0 \cdot 0 - 6 \cdot 0)\big) = 2(2 \cdot 6 \cdot 12) = 288.$$

1 c)

$$6\begin{vmatrix} 2 & 1 & 1 \\ 0 & 6 & 5 \\ 0 & 0 & 4 \end{vmatrix} = 6\big(2(6 \cdot 4 - 5 \cdot 0) - 1(0 \cdot 4 - 5 \cdot 0) + 1(0 \cdot 0 - 6 \cdot 0)\big) = 6(2 \cdot 6 \cdot 4) = 288.$$

1 d)

$$12\begin{vmatrix} 1 & 1 & 1 \\ 0 & 6 & 5 \\ 0 & 0 & 4 \end{vmatrix} = 12\big(1(6 \cdot 4 - 5 \cdot 0) - 1(0 \cdot 4 - 5 \cdot 0) + 1(0 \cdot 0 - 6 \cdot 0)\big) = 12(1 \cdot 6 \cdot 4) = 288.$$

1 e)

$$48\begin{vmatrix} 1 & 1 & 1 \\ 0 & 6 & 5 \\ 0 & 0 & 1 \end{vmatrix} = 48\big(1(6 \cdot 1 - 5 \cdot 0) - 1(0 \cdot 1 - 5 \cdot 0) + 1(0 \cdot 0 - 6 \cdot 0)\big) = 48(1 \cdot 6 \cdot 1) = 288.$$

2 a)

$$|\mathbf{A}| = \begin{vmatrix} 22 & 0 & 0 \\ 660 & 15 & 0 \\ 220 & -12 & 2 \end{vmatrix} = 22(15{\cdot}2 - 0(-12)) - 0(660{\cdot}2 - 0{\cdot}220) + 0(660(-12) - 15{\cdot}220) = 22{\cdot}15{\cdot}2 = 660.$$

2 b)

$$2\begin{vmatrix} 22 & 0 & 0 \\ 660 & 15 & 0 \\ 220 & -12 & 1 \end{vmatrix} = 2\big(22(15{\cdot}1 - 0(-12)) - 0(660{\cdot}1 - 0{\cdot}220) + 0(660(-12) - 15{\cdot}220)\big) = 2(22{\cdot}15{\cdot}1) = 660.$$

2 c)

$$4\begin{vmatrix} 11 & 0 & 0 \\ 330 & 15 & 0 \\ 110 & -12 & 1 \end{vmatrix} = 4\big(11(15\cdot 1-0(-12))-0(330\cdot 1-0\cdot 110)+0(330(-12)-15\cdot 110)\big) = 4(11\cdot 15\cdot 1) = 660.$$

2 d)

$$12\begin{vmatrix} 11 & 0 & 0 \\ 110 & 5 & 0 \\ 110 & -12 & 1 \end{vmatrix} = 12\big(11(5\cdot 1-0(-12))-0(110\cdot 1-0\cdot 110)+0(110(-12)-5\cdot 110)\big) = 12(11\cdot 5\cdot 1) = 660.$$

2 e)

$$60\begin{vmatrix} 11 & 0 & 0 \\ 22 & 1 & 0 \\ 110 & -12 & 1 \end{vmatrix} = 60\big(11(1\cdot 1-0(-12))-0(22\cdot 1-0\cdot 110)+0(22(-12)-1\cdot 110)\big) = 60(11\cdot 1\cdot 1) = 660.$$

2 f)

$$660\begin{vmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 10 & -12 & 1 \end{vmatrix} = 660\big(1(1\cdot 1-0(-12))-0(2\cdot 1-0\cdot 10)+0(2(-12)-1\cdot 10)\big) = 660(1\cdot 1\cdot 1) = 660.$$

2 g)

$$\begin{aligned} |\mathbf{A}| &= \begin{vmatrix} 22 & 0 & 0 \\ 660 & 15 & 0 \\ 220 & -12 & 2 \end{vmatrix} = 2\begin{vmatrix} 22 & 0 & 0 \\ 660 & 15 & 0 \\ 220 & -12 & 1 \end{vmatrix} = 2\cdot 2\begin{vmatrix} 11 & 0 & 0 \\ 330 & 15 & 0 \\ 110 & -12 & 1 \end{vmatrix} \\ &= 2\cdot 2\cdot 3\begin{vmatrix} 11 & 0 & 0 \\ 110 & 5 & 0 \\ 110 & -12 & 1 \end{vmatrix} = 2\cdot 2\cdot 3\cdot 5\begin{vmatrix} 11 & 0 & 0 \\ 22 & 1 & 0 \\ 110 & -12 & 1 \end{vmatrix} \\ &= 2\cdot 2\cdot 3\cdot 5\cdot 11\begin{vmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 10 & -12 & 1 \end{vmatrix} = 2\cdot 2\cdot 3\cdot 5\cdot 11\cdot 1. \\ &\therefore |\mathbf{A}| = 660 = 2^2\cdot 3\cdot 5\cdot 11. \end{aligned}$$

3 a)

$$\begin{aligned} |\mathbf{A}| &= \begin{vmatrix} k & 3k^3 & 4k^3(k-1) \\ 0 & k^2 & 2k^2(k-1) \\ 0 & 0 & 3k^2(k-1) \end{vmatrix} \\ &= k(k^2\cdot 3k^2(k-1)-2k^2(k-1)\cdot 0)-3k^3(0\cdot 3k^2(k-1)-2k^2(k-1)\cdot 0) \\ &\quad +4k^3(k-1)(0\cdot 0-k^2\cdot 0) = k\cdot k^2\cdot 3k^2(k-1) = 3k^6-3k^5 = 3k^5(k-1). \end{aligned}$$

3 b)

$$k\begin{vmatrix}1 & 3k^2 & 4k^2(k-1)\\ 0 & k^2 & 2k^2(k-1)\\ 0 & 0 & 3k^2(k-1)\end{vmatrix} = k(1(k^2\cdot 3k^2(k-1) - 2k^2(k-1)\cdot 0) - 3k^2(0\cdot 3k^2(k-1) - 2k^2(k-1)\cdot 0) + 4k^2(k-1)(0\cdot 0 - k^2\cdot 0)) = 3k^6 - 3k^5 = 3k^5(k-1).$$

3 c)

$$k(k-1)\begin{vmatrix}1 & 3k^2 & 4k^2\\ 0 & k^2 & 2k^2\\ 0 & 0 & 3k^2\end{vmatrix} = k(k-1)(1(k^2\cdot 3k^2 - 2k^2\cdot 0) - 3k^2(0\cdot 3k^2 - 2k^2\cdot 0) + 4k^2(0\cdot 0 - k^2\cdot 0)) = 3k^6 - 3k^5 = 3k^5(k-1).$$

3 d)

$$k^3(k-1)\begin{vmatrix}1 & 3k^2 & 4\\ 0 & k^2 & 2\\ 0 & 0 & 3\end{vmatrix} = k^3(k-1)\big(1(k^2\cdot 3 - 2\cdot 0) - 3k^2(0\cdot 3 - 2\cdot 0) + 4(0\cdot 0 - k^2\cdot 0)\big) = 3k^6 - 3k^5 = 3k^5(k-1).$$

3 e)

$$k^5(k-1)\begin{vmatrix}1 & 3 & 4\\ 0 & 1 & 2\\ 0 & 0 & 3\end{vmatrix} = k^5(k-1)\big(1(1\cdot 3 - 2\cdot 0) - 3(0\cdot 3 - 2\cdot 0) + 4(0\cdot 0 - 1\cdot 0)\big) = 3k^6 - 3k^5 = 3k^5(k-1).$$

3 f)

$|\mathbf{A}| = 3k^5(k-1).$

4 a)

$$|\mathbf{A}| = (5k-10)((6-2k)(3k-18) - 0(5k-15)) - 0((4k-8)(3k-18) - 0(12-6k)) + 0((4k-8)(5k-15) - (6-2k)(12-6k)) = (5k-10)(6-2k)(3k-18).$$

4 b)

$$\begin{aligned}|\mathbf{A}| &= (k-2)\begin{vmatrix}5 & 0 & 0\\ 4 & 6-2k & 0\\ -6 & 5k-15 & 3k-18\end{vmatrix}\\ &= (k-2)(5((6-2k)(3k-18) - 0(5k-15)) - 0(4(3k-18) - 0(-6)) + 0(4(5k-15) - (6-2k)(-6)))\\ &= (k-2)\cdot 5(6-2k)(3k-18) = 5(k-2)(6-2k)(3k-18).\end{aligned}$$

4 c)

$$
\begin{aligned}
|\mathbf{A}| &= (k-2)(k-3)\begin{vmatrix} 5 & 0 & 0 \\ 4 & -2 & 0 \\ -6 & 5 & 3k-18 \end{vmatrix} \\
&= (k-2)(k-3)\big(5((-2)(3k-18)-0\cdot 5)-0(4(3k-18)-0(-6))+0(4\cdot 5-(-2)(-6))\big) \\
&= (k-2)(k-3)\cdot 5(-2)(3k-18) = -10(k-2)(k-3)(3k-18).
\end{aligned}
$$

4 d)

$$
\begin{aligned}
|\mathbf{A}| &= (k-2)(k-3)(k-6)\begin{vmatrix} 5 & 0 & 0 \\ 4 & -2 & 0 \\ -6 & 5 & 3 \end{vmatrix} \\
&= (k-2)(k-3)(k-6)\big(5((-2)3-0\cdot 5)-0(4\cdot 3-0(-6))+0(4\cdot 5-(-2)(-6))\big) \\
&= (k-2)(k-3)(k-6)\cdot 5(-2)3 = -30(k-2)(k-3)(k-6).
\end{aligned}
$$

4 e)

$|\mathbf{A}| = -30(k-2)(k-3)(k-6)$.

5)

$$
\begin{aligned}
|\mathbf{A}| &= \begin{vmatrix} 24k & 4k+4 & 3k-6 \\ -6k & 2k+2 & 2k-4 \\ 12k & -3k-3 & 8-4k \end{vmatrix} \\
&= 6k\begin{vmatrix} 4 & 4(k+1) & 3(k-2) \\ -1 & 2(k+1) & 2(k-2) \\ 2 & -3(k+1) & -4(k-2) \end{vmatrix} \\
&= 6k(k+1)\begin{vmatrix} 4 & 4 & 3(k-2) \\ -1 & 2 & 2(k-2) \\ 2 & -3 & -4(k-2) \end{vmatrix} \\
&= 6k(k+1)(k-2)\begin{vmatrix} 4 & 4 & 3 \\ -1 & 2 & 2 \\ 2 & -3 & -4 \end{vmatrix}. \\
&= 6k(k+1)(k+2)(4(2(-4)-2(-3))-4((-1)(-4)-2\cdot 2)+3((-1)(-3)-2\cdot 2)) \\
&= 6k(k+1)(k+2)(4(-2)-4(0)+3(-1)) \\
&= 6k(k+1)(k+2)(-8-0-3) = -66k(k+1)(k+2).
\end{aligned}
$$

6)

$$
|\mathbf{A}| = 0 \Rightarrow \begin{vmatrix} 4k-2 & 8k-4 & 5-10k \\ 3-9k & 18k-6 & 6k-2 \\ -1-4k & 12k+3 & 20k+5 \end{vmatrix} = 0
$$

$$\Rightarrow \begin{vmatrix} 2(2k-1) & 4(2k-1) & -5(2k-1) \\ -3(3k-1) & 6(3k-1) & 2(3k-1) \\ -(4k+1) & 3(4k+1) & 5(4k+1) \end{vmatrix} = 0$$

$$\Rightarrow (2k-1)\begin{vmatrix} 2 & 4 & -5 \\ -3(3k-1) & 6(3k-1) & 2(3k-1) \\ -(4k+1) & 3(4k+1) & 5(4k+1) \end{vmatrix} = 0$$

$$\Rightarrow (2k-1)(3k-1)\begin{vmatrix} 2 & 4 & -5 \\ -3 & 6 & 2 \\ -(4k+1) & 3(4k+1) & 5(4k+1) \end{vmatrix} = 0$$

$$\Rightarrow (2k-1)(3k-1)(4k+1)\begin{vmatrix} 2 & 4 & -5 \\ -3 & 6 & 2 \\ -1 & 3 & 5 \end{vmatrix} = 0$$

$$\Rightarrow (2k-1)(3k-1)(4k+1)(2(6\cdot 5 - 2\cdot 3) - 4((-3)5 - 2(-1)) - 5((-3)3 - 6(-1))) = 0$$

$$\Rightarrow (2k-1)(3k-1)(4k+1)(2(24) - 4(-13) - 5(-3)) = 0$$

$$\Rightarrow 115(2k-1)(3k-1)(4k+1) = 0$$

$$\Rightarrow k = \frac{1}{2} \text{ or } k = \frac{1}{3} \text{ or } k = -\frac{1}{4}.$$

7 a)

$$|\mathbf{A}| = 0(2(-3) - 3\cdot 1) - (-1)(2(-3) - 3\cdot 2) + 3(2\cdot 1 - 2\cdot 2) = 0 - (-1)(-12) + 3(-2) = -18.$$

7 b)

$$|\mathbf{A}| = \begin{vmatrix} 0 & -1 & 3 \\ 2 & 2 & 3 \\ 2 & 1 & -3 \end{vmatrix}.$$

Let the new $R_1 = R_1 + R_3$:

$$|\mathbf{A}| = \begin{vmatrix} 2 & 0 & 0 \\ 2 & 2 & 3 \\ 2 & 1 & -3 \end{vmatrix}.$$

Let the new $R_2 = R_2 + R_3$:

$$|\mathbf{A}| = \begin{vmatrix} 2 & 0 & 0 \\ 4 & 3 & 0 \\ 2 & 1 & -3 \end{vmatrix}.$$

7 c)

$$|\mathbf{A}| = \begin{vmatrix} 2 & 0 & 0 \\ 4 & 3 & 0 \\ 2 & 1 & -3 \end{vmatrix} = 2\cdot 3\cdot(-3) = -18.$$

8 a)

$$|\mathbf{A}| = \begin{vmatrix} 2 & 2 & -1 \\ 2 & 5 & 0 \\ 2 & 5 & 4 \end{vmatrix} = 2(5\cdot 4 - 0\cdot 5) - 2(2\cdot 4 - 0\cdot 2) - 1(2\cdot 5 - 5\cdot 2) = 2(20) - 2(8) - 0 = 24.$$

8 b)

$$|\mathbf{A}| = \begin{vmatrix} 2 & 2 & -1 \\ 2 & 5 & 0 \\ 2 & 5 & 4 \end{vmatrix}.$$

Factor 2 out of C_1:

$$|\mathbf{A}| = 2\begin{vmatrix} 1 & 2 & -1 \\ 1 & 5 & 0 \\ 1 & 5 & 4 \end{vmatrix}.$$

Let the new $R_3 = R_3 - R_2$:

$$|\mathbf{A}| = 2\begin{vmatrix} 1 & 2 & -1 \\ 1 & 5 & 0 \\ 0 & 0 & 4 \end{vmatrix}.$$

Let the new $R_2 = R_2 - R_1$:

$$|\mathbf{A}| = 2\begin{vmatrix} 1 & 2 & -1 \\ 0 & 3 & 1 \\ 0 & 0 & 4 \end{vmatrix}.$$

8 c)

$$|\mathbf{A}| = 2\begin{vmatrix} 1 & 2 & -1 \\ 0 & 3 & 1 \\ 0 & 0 & 4 \end{vmatrix} = 2 \cdot 1 \cdot 3 \cdot 4 = 24.$$

9 a)

$$\begin{vmatrix} 1 & 0 & 5 \\ 2 & 1 & 2 \\ 3 & 2 & 1 \end{vmatrix} \xrightarrow{R_2=R_2-2R_1} \begin{vmatrix} 1 & 0 & 5 \\ 0 & 1 & -8 \\ 3 & 2 & 1 \end{vmatrix} \xrightarrow{R_3=R_3-3R_1} \begin{vmatrix} 1 & 0 & 5 \\ 0 & 1 & -8 \\ 0 & 2 & -14 \end{vmatrix}$$

$$\xrightarrow{R_3=R_3-2R_2} \begin{vmatrix} 1 & 0 & 5 \\ 0 & 1 & -8 \\ 0 & 0 & 2 \end{vmatrix} = 1 \cdot 1 \cdot 2 = 2.$$

9 b)

$$\begin{vmatrix} 2 & 4 & 6 \\ 3 & -1 & 1 \\ -2 & 2 & 1 \end{vmatrix} = 2\begin{vmatrix} 1 & 2 & 3 \\ 3 & -1 & 1 \\ -2 & 2 & 1 \end{vmatrix} \xrightarrow{R_2=R_2-3R_1} 2\begin{vmatrix} 1 & 2 & 3 \\ 0 & -7 & -8 \\ -2 & 2 & 1 \end{vmatrix}$$

$$\xrightarrow{R_3=R_3+2R_1} 2\begin{vmatrix} 1 & 2 & 3 \\ 0 & -7 & -8 \\ 0 & 6 & 7 \end{vmatrix} = 14\begin{vmatrix} 1 & 2 & 3 \\ 0 & -1 & -\frac{8}{7} \\ 0 & 6 & 7 \end{vmatrix}$$

$$\xrightarrow{R_3=R_3+6R_2} 14\begin{vmatrix} 1 & 2 & 3 \\ 0 & -1 & -\frac{8}{7} \\ 0 & 0 & \frac{1}{7} \end{vmatrix} = 14(1 \cdot (-1) \cdot \frac{1}{7}) = -2.$$

9 c)

$$\begin{vmatrix} 4 & 1 & 6 \\ -1 & 3 & 3 \\ 4 & -2 & 3 \end{vmatrix} \xrightarrow{R_3=R_3-4R_1} \begin{vmatrix} 4 & 1 & 6 \\ -1 & 3 & 3 \\ 0 & -3 & -3 \end{vmatrix} = 4 \begin{vmatrix} 1 & \frac{1}{4} & \frac{3}{2} \\ -1 & 3 & 3 \\ 0 & -3 & -3 \end{vmatrix}$$

$$\xrightarrow{R_2=R_2+R_1} 4 \begin{vmatrix} 1 & \frac{1}{4} & \frac{3}{2} \\ 0 & \frac{13}{4} & \frac{9}{2} \\ 0 & -3 & -3 \end{vmatrix} = 13 \begin{vmatrix} 1 & \frac{1}{4} & \frac{3}{2} \\ 0 & 1 & \frac{18}{13} \\ 0 & -3 & -3 \end{vmatrix}$$

$$\xrightarrow{R_3=R_3+3R_2} 13 \begin{vmatrix} 1 & \frac{1}{4} & \frac{3}{2} \\ 0 & 1 & \frac{18}{13} \\ 0 & 0 & \frac{15}{13} \end{vmatrix} = 13 \cdot 1 \cdot 1 \cdot \frac{15}{13} = 15.$$

9 d)

$$\begin{vmatrix} 3 & -2 & 5 \\ -2 & 4 & 2 \\ -4 & -1 & 9 \end{vmatrix} = 3 \begin{vmatrix} 1 & -\frac{2}{3} & \frac{5}{3} \\ -2 & 4 & 2 \\ -4 & -1 & 9 \end{vmatrix} \xrightarrow{R_2=R_2+2R_1} 3 \begin{vmatrix} 1 & -\frac{2}{3} & \frac{5}{3} \\ 0 & \frac{8}{3} & \frac{16}{3} \\ -4 & -1 & 9 \end{vmatrix}$$

$$\xrightarrow{R_3=R_3+4R_1} 3 \begin{vmatrix} 1 & -\frac{2}{3} & \frac{5}{3} \\ 0 & \frac{8}{3} & \frac{16}{3} \\ 0 & -\frac{11}{3} & \frac{47}{3} \end{vmatrix} = 8 \begin{vmatrix} 1 & -\frac{2}{3} & \frac{5}{3} \\ 0 & 1 & 2 \\ 0 & -\frac{11}{3} & \frac{47}{3} \end{vmatrix}$$

$$\xrightarrow{R_3=R_3+\frac{11}{3}R_2} 8 \begin{vmatrix} 1 & -\frac{2}{3} & \frac{5}{3} \\ 0 & 1 & 2 \\ 0 & 0 & 23 \end{vmatrix} = 8(1 \cdot 1 \cdot 23) = 184.$$

10 a)

$$|\mathbf{A}| = \begin{vmatrix} 2 & 1 & 3 \\ 0 & -2 & 3 \\ 0 & 0 & 5 \end{vmatrix} = 2((-2)5 - 3 \cdot 0) - (0 \cdot 5 - 3 \cdot 0) + 3(0 \cdot 0 - (-2) \cdot 0) = 2(-2)5 = -20.$$

10 b)

$$\begin{vmatrix} 0 & 0 & 5 \\ 2 & 1 & 3 \\ 0 & -2 & 3 \end{vmatrix} = 0(1 \cdot 3 - 3(-2)) - 0(2 \cdot 3 - 3 \cdot 0) + 5(2(-2) - 1 \cdot 0) = 5 \cdot 2(-2) = -20.$$

10 c)

$$\begin{vmatrix} 0 & -2 & 3 \\ 0 & 0 & 5 \\ 2 & 1 & 3 \end{vmatrix} = 0(0 \cdot 3 - 5 \cdot 1) + 2(0 \cdot 3 - 5 \cdot 2) + 3(0 \cdot 1 - 0 \cdot 2) = 2(-5 \cdot 2) = -20.$$

10 d)

$$\begin{vmatrix} 1 & 3 & 2 \\ -2 & 3 & 0 \\ 0 & 5 & 0 \end{vmatrix} = (3 \cdot 0 - 0 \cdot 5) - 3((-2) \cdot 0 - 0 \cdot 0) + 2((-2)5 - 3 \cdot 0) = 2(-2)5 = -20.$$

10 e)

$$\begin{vmatrix} 3 & 2 & 1 \\ 3 & 0 & -2 \\ 5 & 0 & 0 \end{vmatrix} = 3(0 \cdot 0 - (-2) \cdot 0) - 2(3 \cdot 0 - (-2)5) + (2 \cdot 0 - 0 \cdot 5) = -2(-(-2)5) = -20.$$

10 f)

$$\begin{vmatrix} 0 & -2 & 3 \\ 2 & 1 & 3 \\ 0 & 0 & 5 \end{vmatrix} = 0(1 \cdot 5 - 3 \cdot 0) + 2(2 \cdot 5 - 3 \cdot 0) + 3(2 \cdot 0 - 1 \cdot 0) = 2 \cdot 2 \cdot 5 = 20.$$

10 g)

$$\begin{vmatrix} 1 & 2 & 3 \\ -2 & 0 & 3 \\ 0 & 0 & 5 \end{vmatrix} = (0 \cdot 5 - 3 \cdot 0) - 2((-2)5 - 3 \cdot 0) + 3((-2) \cdot 0 - 0 \cdot 0) = -2(-2)5 = 20.$$

10 h)

If you rotate the rows or columns, the determinant does not change. If you swap any two rows or columns, then the determinant changes sign.

11)

R_1 and R_3 have been swapped, so the value is -6.

12 a)

$$\begin{vmatrix} 3 & 0 & 0 \\ 0 & 0 & 2 \\ 0 & 5 & 0 \end{vmatrix} \xrightarrow{R_2 \leftrightarrow R_3} - \begin{vmatrix} 3 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 2 \end{vmatrix} = -3 \cdot 5 \cdot 2 = -30.$$

12 b)

$$\begin{vmatrix} 0 & 0 & 2 \\ 0 & 4 & 0 \\ -5 & 0 & 0 \end{vmatrix} \xrightarrow{R_1 \leftrightarrow R_3} - \begin{vmatrix} -5 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 2 \end{vmatrix} = -(-5) \cdot 4 \cdot 2 = 40.$$

12 c)

$$\begin{vmatrix} 0 & 0 & 6k \\ k & 0 & 0 \\ 0 & 4k & 0 \end{vmatrix} \xrightarrow{\text{rotate all rows up one position}} \begin{vmatrix} k & 0 & 0 \\ 0 & 4k & 0 \\ 0 & 0 & 6k \end{vmatrix} = k \cdot 4k \cdot 6k = 24k^3.$$

12 d)

$$\begin{vmatrix} 0 & pq & 0 \\ p+q & 0 & 0 \\ 0 & 0 & pq \end{vmatrix} \xrightarrow{C_1 \leftrightarrow C_2} - \begin{vmatrix} pq & 0 & 0 \\ 0 & p+q & 0 \\ 0 & 0 & pq \end{vmatrix} = -pq(p+q)pq = -p^2q^2(p+q).$$

13 a)

$$\begin{vmatrix} 1 & 1 & 1 \\ 3 & 4 & -5 \\ 1 & 1 & 1 \end{vmatrix} \xrightarrow{R_1=R_1-R_3} \begin{vmatrix} 0 & 0 & 0 \\ 3 & 4 & -5 \\ 1 & 1 & 1 \end{vmatrix} = 0 \quad \text{as all of the entries in a single row are zero.}$$

13 b)

$$\begin{vmatrix} 2 & -3 & 4 \\ 3 & 0 & 6 \\ 4 & 2 & 8 \end{vmatrix} \xrightarrow{C_1=C_1-\frac{1}{2}C_3} \begin{vmatrix} 0 & -3 & 4 \\ 0 & 0 & 6 \\ 0 & 2 & 8 \end{vmatrix} = 0 \quad \text{as all of the entries in a single column are zero.}$$

14 a)

$$\begin{vmatrix} 4k+1 & k & 4k+1 \\ 3k & k & 3k \\ -2 & 4 & 2k+2 \end{vmatrix} \xrightarrow{R_1=R_1-R_2} \begin{vmatrix} k+1 & 0 & k+1 \\ 3k & k & 3k \\ -2 & 4 & 2k+2 \end{vmatrix} \xrightarrow{C_3=C_3-C_1} \begin{vmatrix} k+1 & 0 & 0 \\ 3k & k & 0 \\ -2 & 4 & 2k+4 \end{vmatrix}.$$

14 b)

$|\mathbf{A}| = (k+1) \cdot k \cdot (2k+4) = 2k(k+1)(k+2)$.

15 a)

$$\begin{vmatrix} 1 & p & -p \\ 1 & p+2 & 1 \\ -1 & -p & p+1 \end{vmatrix} \xrightarrow{R_3=R_3+R_1} \begin{vmatrix} 1 & p & -p \\ 1 & p+2 & 1 \\ 0 & 0 & 1 \end{vmatrix} \xrightarrow{R_2=R_2-R_1} \begin{vmatrix} 1 & p & -p \\ 0 & 2 & 1+p \\ 0 & 0 & 1 \end{vmatrix} = 1 \cdot 2 \cdot 1 = 2.$$

15 b)

$$\begin{vmatrix} p-1 & 1 & p-1 \\ p-2 & 1 & p-2 \\ 3p-3 & p & 3p \end{vmatrix} \xrightarrow{R_1=R_1-R_2} \begin{vmatrix} 1 & 0 & 1 \\ p-2 & 1 & p-2 \\ 3p-3 & p & 3p \end{vmatrix} \xrightarrow{C_3=C_3-C_1} \begin{vmatrix} 1 & 0 & 0 \\ p-2 & 1 & 0 \\ 3p-3 & p & 3 \end{vmatrix} = 1 \cdot 1 \cdot 3 = 3.$$

15 c)

$$\begin{vmatrix} 2 & -p & -1 \\ 1 & 2p-5 & 2 \\ p & 2 & 1 \end{vmatrix} \xrightarrow{R_2=R_2+2R_1} \begin{vmatrix} 2 & -p & -1 \\ 5 & -5 & 0 \\ p & 2 & 1 \end{vmatrix} \xrightarrow{R_3=R_3+R_1} \begin{vmatrix} 2 & -p & -1 \\ 5 & -5 & 0 \\ p+2 & 2-p & 0 \end{vmatrix}$$

$$\xrightarrow{C_2=C_2+C_1} \begin{vmatrix} 2 & 2-p & -1 \\ 5 & 0 & 0 \\ p+2 & 4 & 0 \end{vmatrix} = (-1)(5 \cdot 4) = -20.$$

15 d)

$$\begin{vmatrix} p-2 & 3p-3 & p-1 \\ p+3 & p+14 & p+3 \\ p-1 & 3p-2 & p \end{vmatrix} \xrightarrow{R_3=R_3-R_1} \begin{vmatrix} p-2 & 3p-3 & p-1 \\ p+3 & p+14 & p+3 \\ 1 & 1 & 1 \end{vmatrix} \xrightarrow{C_2=C_2-C_1} \begin{vmatrix} p-2 & 2p-1 & p-1 \\ p+3 & 11 & p+3 \\ 1 & 0 & 1 \end{vmatrix}$$

$$\xrightarrow{C_3=C_3-C_1} \begin{vmatrix} p-2 & 2p-1 & 1 \\ p+3 & 11 & 0 \\ 1 & 0 & 0 \end{vmatrix} = 1 \cdot (-1) \cdot 11 = -11.$$

16 a)

$$\begin{vmatrix} n+5 & 2n+4 & 9-n \\ 4-n & 4-n & 8-3n \\ n+3 & 2n+2 & 5 \end{vmatrix} \xrightarrow{C_1=C_1-C_2} \begin{vmatrix} 1-n & 2n+4 & 9-n \\ 0 & 4-n & 8-3n \\ 1-n & 2n+2 & 5 \end{vmatrix}$$

$$= (1-n)\begin{vmatrix} 1 & 2n+4 & 9-n \\ 0 & 4-n & 8-3n \\ 1 & 2n+2 & 5 \end{vmatrix} \xrightarrow{R_3=R_3-R_1} (1-n)\begin{vmatrix} 1 & 2n+4 & 9-n \\ 0 & 4-n & 8-3n \\ 0 & -2 & n-4 \end{vmatrix}$$

$$\xrightarrow{R_2=R_2+2R_3} (1-n)\begin{vmatrix} 1 & 2n+4 & 9-n \\ 0 & -n & -n \\ 0 & -2 & n-4 \end{vmatrix}$$

$$= -n(1-n)\begin{vmatrix} 1 & 2n+4 & 9-n \\ 0 & 1 & 1 \\ 0 & -2 & n-4 \end{vmatrix} \xrightarrow{R_3=R_3+2R_2} -n(1-n)\begin{vmatrix} 1 & 2n+4 & 9-n \\ 0 & 1 & 1 \\ 0 & 0 & n-2 \end{vmatrix}$$

$$= -n(1-n)(n-2) = n(n-1)(n-2).$$

16 b)

$$\begin{vmatrix} -2n & 1-3n & 1 \\ -4-n & 5-2n & 5 \\ -1-3n & 2-2n & n+2 \end{vmatrix} \xrightarrow{C_2=C_2-C_3} \begin{vmatrix} -2n & -3n & 1 \\ -4-n & -2n & 5 \\ -1-3n & -3n & n+2 \end{vmatrix} = n\begin{vmatrix} -2n & -3 & 1 \\ -4-n & -2 & 5 \\ -1-3n & -3 & n+2 \end{vmatrix}$$

$$= n\begin{vmatrix} -3 & 1 & -2n \\ -2 & 5 & -4-n \\ -3 & n+2 & -1-3n \end{vmatrix} \xrightarrow{C_3=C_3-C_1} n\begin{vmatrix} -3 & 1 & -2n \\ -2 & 5 & -4-n \\ 0 & n+1 & -1-n \end{vmatrix}$$

$$= n(n+1)\begin{vmatrix} -3 & 1 & -2n \\ -2 & 5 & -4-n \\ 0 & 1 & -1 \end{vmatrix} = \frac{1}{2}n(n+1)\begin{vmatrix} -6 & 2 & -4n \\ -2 & 5 & -4-n \\ 0 & 1 & -1 \end{vmatrix}$$

$$= \frac{1}{6}n(n+1)\begin{vmatrix} -6 & 2 & -4n \\ -6 & 15 & -12-3n \\ 0 & 1 & -1 \end{vmatrix} \xrightarrow{R_2=R_2-R_1} \frac{1}{6}n(n+1)\begin{vmatrix} -6 & 2 & -4n \\ 0 & 13 & n-12 \\ 0 & 1 & -1 \end{vmatrix}$$

$$\xrightarrow{R_2=R_2-13R_3} \frac{1}{6}n(n+1)\begin{vmatrix} -6 & 2 & -4n \\ 0 & 0 & n+1 \\ 0 & 1 & -1 \end{vmatrix} \xrightarrow{R_2\leftrightarrow R_3} -\frac{1}{6}n(n+1)\begin{vmatrix} -6 & 2 & -4n \\ 0 & 1 & -1 \\ 0 & 0 & n+1 \end{vmatrix}$$

$$= -\frac{1}{6}n(n+1)(-6)(n+1) = n(n+1)^2.$$

16 c)

$$\begin{vmatrix} 2n-4 & 3n-6 & 4-2n \\ -6 & 4n-6 & 2-5n \\ 3n-6 & 3n-6 & 2-n \end{vmatrix} = \begin{vmatrix} 2(n-2) & 3(n-2) & -2(n-2) \\ -6 & 4n-6 & 2-5n \\ 3(n-2) & 3(n-2) & -(n-2) \end{vmatrix}$$

$$\xrightarrow{R_1=(n-2)R_1} (n-2)\begin{vmatrix} 2 & 3 & -2 \\ -6 & 4n-6 & 2-5n \\ 3(n-2) & 3(n-2) & -(n-2) \end{vmatrix}$$

$$\xrightarrow{R_3=(n-2)R_3} (n-2)^2\begin{vmatrix} 2 & 3 & -2 \\ -6 & 4n-6 & 2-5n \\ 3 & 3 & -1 \end{vmatrix}$$

$$\xrightarrow{R_2=R_2+2R_3} (n-2)^2\begin{vmatrix} 2 & 3 & -2 \\ 0 & 4n & -5n \\ 3 & 3 & -1 \end{vmatrix}$$

$$= n(n-2)^2\begin{vmatrix} 2 & 3 & -2 \\ 0 & 4 & -5 \\ 3 & 3 & -1 \end{vmatrix} = 2n(n-2)^2\begin{vmatrix} 1 & \frac{3}{2} & -1 \\ 0 & 4 & -5 \\ 3 & 3 & -1 \end{vmatrix}$$

$$\xrightarrow{R_3=R_3-3R_1} 2n(n-2)^2\begin{vmatrix} 1 & \frac{3}{2} & -1 \\ 0 & 4 & -5 \\ 0 & -\frac{3}{2} & 2 \end{vmatrix}$$

$$= 8n(n-2)^2\begin{vmatrix} 1 & \frac{3}{2} & -1 \\ 0 & 1 & -\frac{5}{4} \\ 0 & -\frac{3}{2} & 2 \end{vmatrix} \xrightarrow{R_3=R_3+\frac{3}{2}R_2} 8n(n-2)^2\begin{vmatrix} 1 & \frac{3}{2} \\ 0 & 1 \\ 0 & 0 \end{vmatrix}$$

$$= 8n(n-2)^2 \cdot \frac{1}{8} = n(n-2)^2.$$

16 d)

$$\begin{vmatrix} 9-15n & 12-17n & 12-15n \\ 40-49n & 40-49n & 40-50n \\ 13-20n & 16-22n & 16-20n \end{vmatrix} = \begin{vmatrix} 9-15n & 12-17n & 3(4-5n) \\ 40-49n & 40-49n & 10(4-5n) \\ 13-20n & 16-22n & 4(4-5n) \end{vmatrix}$$

$$= (4-5n)\begin{vmatrix} 9-15n & 12-17n & 3 \\ 40-49n & 40-49n & 10 \\ 13-20n & 16-22n & 4 \end{vmatrix} \xrightarrow{C_1=C_1-C_2} (4-5n)\begin{vmatrix} 2n-3 & 12-17n & 3 \\ 0 & 40-49n & 10 \\ 2n-3 & 16-22n & 4 \end{vmatrix}$$

$$= (2n-3)(4-5n)\begin{vmatrix} 1 & 12-17n & 3 \\ 0 & 40-49n & 10 \\ 1 & 16-22n & 4 \end{vmatrix} \xrightarrow{R_3=R_3-R_1} (2n-3)(4-5n)\begin{vmatrix} 1 & 12-17n \\ 0 & 40-49n \\ 0 & 4-5n \end{vmatrix}$$

$$\xrightarrow{R_2=R_2-10R_3} (2n-3)(4-5n)\begin{vmatrix} 1 & 12-17n & 3 \\ 0 & n & 0 \\ 0 & 4-5n & 1 \end{vmatrix}$$

$$= n(2n-3)(4-5n)\begin{vmatrix} 1 & 12-17n & 3 \\ 0 & 1 & 0 \\ 0 & 4-5n & 1 \end{vmatrix} = n(2n-3)(4-5n).$$

17 a)

At stage (3), the a_{33} element (bottom right) should be $4n-1$ as $(n-3)+(3n+2)=4n-1$.

17 b)

$$\begin{vmatrix} n-1 & 2 & -1 \\ -n & n & n \\ -2n-1 & 3n+2 & 4n+1 \end{vmatrix} = n \begin{vmatrix} n-1 & 2 & -1 \\ -1 & 1 & 1 \\ -2n-1 & 3n+2 & 4n+1 \end{vmatrix} \xrightarrow{C_1=C_1+C_2} n \begin{vmatrix} n+1 & 2 & -1 \\ 0 & 1 & 1 \\ n+1 & 3n+2 & 4n+1 \end{vmatrix}$$

$$= n(n+1) \begin{vmatrix} 1 & 2 & -1 \\ 0 & 1 & 1 \\ 1 & 3n+2 & 4n+1 \end{vmatrix} \xrightarrow{R_3=R_3-R_1} n(n+1) \begin{vmatrix} 1 & 2 & -1 \\ 0 & 1 & 1 \\ 0 & 3n & 4n+2 \end{vmatrix}$$

$$\xrightarrow{C_3=C_3-C_2} n(n+1) \begin{vmatrix} 1 & 2 & -3 \\ 0 & 1 & 0 \\ 0 & 3n & n+2 \end{vmatrix} = n(n+1)(n+2).$$

18 a)

$$\begin{vmatrix} n+1 & n^2 & 1 \\ -n-1 & n+1-n^2 & n \\ n+1 & 2n^2-n & 2n^2-n \end{vmatrix} = \begin{vmatrix} n+1 & n^2 & 1 \\ -(n+1) & n+1-n^2 & n \\ n+1 & 2n^2-n & 2n^2-n \end{vmatrix}$$

$$= (n+1) \begin{vmatrix} 1 & n^2 & 1 \\ -1 & n+1-n^2 & n \\ 1 & 2n^2-n & 2n^2-n \end{vmatrix} \xrightarrow{R_3=R_3-R_1} (n+1) \begin{vmatrix} 1 & n^2 & 1 \\ -1 & n+1-n^2 & n \\ 0 & n^2-n & 2n^2- \end{vmatrix}$$

$$\xrightarrow{R_2=R_2+R_1} (n+1) \begin{vmatrix} 1 & n^2 & 1 \\ 0 & n+1 & n+1 \\ 0 & n^2-n & 2n^2-n-1 \end{vmatrix}$$

$$= (n+1)^2 \begin{vmatrix} 1 & n^2 & 1 \\ 0 & 1 & 1 \\ 0 & n^2-n & 2n^2-n-1 \end{vmatrix} = (n+1)^2 \begin{vmatrix} 1 & n^2 & 1 \\ 0 & 1 & 1 \\ 0 & n(n-1) & (2n+1)(n-1) \end{vmatrix}$$

$$= (n+1)^2(n-1) \begin{vmatrix} 1 & n^2 & 1 \\ 0 & 1 & 1 \\ 0 & n & 2n+1 \end{vmatrix} \xrightarrow{C_3=C_3-C_2} (n+1)^2(n-1) \begin{vmatrix} 1 & n^2 & 1-n^2 \\ 0 & 1 & 0 \\ 0 & n & n+1 \end{vmatrix}$$

$$= (n+1)^3(n-1) \begin{vmatrix} 1 & n^2 & 1-n \\ 0 & 1 & 0 \\ 0 & n & 1 \end{vmatrix} = (n+1)^3(n-1).$$

18 b)

$$\begin{vmatrix} 3n-2 & n^2+2n & 2n-1 \\ 3n^2-1 & n+2 & 3n^2-1 \\ n^2+4n-4 & n^2+2n & n^2+3n-3 \end{vmatrix} = \begin{vmatrix} 3n-2 & n(n+2) & 2n-1 \\ 3n^2-1 & n+2 & 3n^2-1 \\ n^2+4n-4 & n(n+2) & n^2+3n-3 \end{vmatrix}$$

$$= (n+2) \begin{vmatrix} 3n-2 & n & 2n-1 \\ 3n^2-1 & 1 & 3n^2-1 \\ n^2+4n-4 & n & n^2+3n-3 \end{vmatrix} \xrightarrow{C_1=C_1-C_3} (n+2) \begin{vmatrix} n-1 & n & 2 \\ 0 & 1 & 3n \\ n-1 & n & n^2+ \end{vmatrix}$$

$$= (n-1)(n+2)\begin{vmatrix} 1 & n & 2n-1 \\ 0 & 1 & 3n^2-1 \\ 1 & n & n^2+3n-3 \end{vmatrix} \xrightarrow{R_3=R_3-R_1} (n-1)(n+2)\begin{vmatrix} 1 & n & 2n \\ 0 & 1 & 3n \\ 0 & 0 & n^2+ \end{vmatrix}$$

$$= (n-1)(n+2)\begin{vmatrix} 1 & n & 2n-1 \\ 0 & 1 & 3n^2-1 \\ 0 & 0 & (n+2)(n-1) \end{vmatrix} = (n-1)^2(n+2)^2.$$

18 c)

$$\begin{vmatrix} 3n-n^2 & n+1 & n^2-n-6 \\ 6+n-n^2 & -n^2-n-2 & n^2-9 \\ 6-2n & -n^2-3n & n-3 \end{vmatrix} = \begin{vmatrix} n(3-n) & n+1 & (n-3)(n+2) \\ (n+2)(3-n) & -n^2-n-2 & (n-3)(n+3) \\ 2(3-n) & -n^2-3n & n-3 \end{vmatrix}$$

$$= (3-n)\begin{vmatrix} n & n+1 & (n-3)(n+2) \\ n+2 & -n^2-n-2 & (n-3)(n+3) \\ 2 & -n^2-3n & n-3 \end{vmatrix}$$

$$= (3-n)(n-3)\begin{vmatrix} n & n+1 & n+2 \\ n+2 & -n^2-n-2 & n+3 \\ 2 & -n^2-3n & 1 \end{vmatrix}$$

$$= -(n-3)^2\begin{vmatrix} n & n+1 & n+2 \\ n+2 & -n^2-n-2 & n+3 \\ 2 & -n^2-3n & 1 \end{vmatrix}$$

$$\xrightarrow{R_2=R_2-R_1} -(n-3)^2\begin{vmatrix} n & n+1 & n+2 \\ 2 & -n^2-2n-3 & 1 \\ 2 & -n^2-3n & 1 \end{vmatrix}$$

$$\xrightarrow{R_3=R_3-R_2} -(n-3)^2\begin{vmatrix} n & n+1 & n+2 \\ 2 & -n^2-2n-3 & 1 \\ 0 & 3-n & 0 \end{vmatrix}$$

$$= (n-3)^3\begin{vmatrix} n & n+1 \\ 2 & -n^2-2n-3 \\ 0 & 1 \end{vmatrix}$$

$$\xrightarrow{C_1=C_1-C_3} (n-3)^3\begin{vmatrix} -2 & n+1 & n+2 \\ 1 & -n^2-2n-3 & 1 \\ 0 & 1 & 0 \end{vmatrix}$$

$$\xrightarrow{R_1=R_1+2R_2} (n-3)^3\begin{vmatrix} 0 & -2n^2-3n-5 & n+4 \\ 1 & -n^2-2n-3 & 1 \\ 0 & 1 & 0 \end{vmatrix}$$

$$\xrightarrow{\text{rotate rows one place upwards}} (n-3)^3\begin{vmatrix} 1 & -n^2-2n-3 & 1 \\ 0 & 1 & 0 \\ 0 & -2n^2-3n-5 & n+4 \end{vmatrix} = (n-3)^3($$

18 d)

$$\begin{vmatrix} n-7 & n-12 & n-10 \\ 5n^2-20n & 11n^2-44n & 10n^2-40n \\ 3n-6 & 5n-10 & 5n-8 \end{vmatrix} = \begin{vmatrix} n-7 & n-12 & n-10 \\ 5n(n-4) & 11n(n-4) & 10n(n-4) \\ 3n-6 & 5n-10 & 5n-8 \end{vmatrix}$$

$$= n(n-4)\begin{vmatrix} n-7 & n-12 & n-10 \\ 5 & 11 & 10 \\ 3n-6 & 5n-10 & 5n-8 \end{vmatrix} \xrightarrow{R_3=R_3-R_1} n(n-4)\begin{vmatrix} n-7 & n-12 \\ 5 & 11 \\ 2n+1 & 4n+2 \end{vmatrix}$$

$$= n(n-4)\begin{vmatrix} n-7 & n-12 & n-10 \\ 5 & 11 & 10 \\ 2n+1 & 2(2n+1) & 2(2n+1) \end{vmatrix}$$

$$= n(n-4)(2n+1)\begin{vmatrix} n-7 & n-12 & n-10 \\ 5 & 11 & 10 \\ 1 & 2 & 2 \end{vmatrix} \xrightarrow{C_1=C_1-C_2} n(n-4)(2n+1)\begin{vmatrix}$$

$$= 5n(n-4)(2n+1)\begin{vmatrix} 1 & \frac{1}{5}n-\frac{12}{5} & \frac{1}{5}n-2 \\ -6 & 11 & 10 \\ -1 & 2 & 2 \end{vmatrix} \xrightarrow{R_2=R_2+6R_1} 5n(n-4)(2n+1$$

$$\xrightarrow{R_3=R_3+R_1} 5n(n-4)(2n+1)\begin{vmatrix} 1 & \frac{1}{5}n-\frac{12}{5} & \frac{1}{5}n-2 \\ 0 & \frac{6}{5}n-\frac{17}{5} & \frac{6}{5}n-2 \\ 0 & \frac{1}{5}n-\frac{2}{5} & \frac{1}{5}n \end{vmatrix}$$

$$\xrightarrow{C_2=C_2-C_3} 5n(n-4)(2n+1)\begin{vmatrix} 1 & - \\ 0 & - \\ 0 & - \end{vmatrix}$$

$$= -7n(n-4)(2n+1)\begin{vmatrix} 1 & -\frac{2}{5} & \frac{1}{5}n-2 \\ 0 & 1 & \frac{6}{7}n-\frac{10}{7} \\ 0 & -\frac{2}{5} & \frac{1}{5}n \end{vmatrix}$$

$$\xrightarrow{R_3=R_3-\frac{2}{5}R_2} -7n(n-4)(2n+1)\begin{vmatrix} 1 & -\frac{2}{5} & \frac{1}{5}n-2 \\ 0 & 1 & \frac{6}{7}n-\frac{10}{7} \\ 0 & 0 & -\frac{1}{7}(n-4) \end{vmatrix}$$

$$= -7n(n-4)(2n+1)\left(-\frac{1}{7}(n-4)\right) = n(n-4)^2(2n+1).$$

19)

$$|\mathbf{A}| = \begin{vmatrix} 5p+q & -2q & 3p-3q \\ 9p & 9q & 6p-6q \\ 3pq+3q^2 & -3q^2 & pq-q^2 \end{vmatrix} = q\begin{vmatrix} 5p+q & -2 & 3p-3q \\ 9p & 9 & 6p-6q \\ 3pq+3q^2 & -3q & pq-q^2 \end{vmatrix}$$

$$\xrightarrow{C_1=C_1-C_3} q\begin{vmatrix} 2p+4q & -2 & 3p-3q \\ 3p+6q & 9 & 6p-6q \\ 2pq+4q^2 & -3q & pq-q^2 \end{vmatrix}$$

$$= q\begin{vmatrix} 2(p+2q) & -2 & 3(p-q) \\ 3(p+2q) & 9 & 6(p-q) \\ 2q(p+2q) & -3q & q(p-q) \end{vmatrix}$$

$$= q(p+2q)\begin{vmatrix} 2 & -2 & 3(p-q) \\ 3 & 9 & 6(p-q) \\ 2q & -3q & q(p-q) \end{vmatrix} = q(p+2q)(p-q)\begin{vmatrix} 2 & -2 & 3 \\ 3 & 9 & 6 \\ 2q & -3q & q \end{vmatrix}$$

$$= q^2(p+2q)(p-q)\begin{vmatrix} 2 & -2 & 3 \\ 3 & 9 & 6 \\ 2 & -3 & 1 \end{vmatrix}$$

$$= q^2(p+2q)(p-q)(2(9\cdot 1 - 6(-3)) + 2(3\cdot 1 - 6\cdot 2) + 3(3(-3) - 9\cdot 2))$$

$$= q^2(p+2q)(p-q)(2\cdot 27 + 2(-9) + 3(-27)) = -45q^2(p+2q)(p-q)$$

$$= 45q^2(p+2q)(q-p).$$

20 a)

$$\begin{vmatrix} 1 & p & pqr \\ 2 & q & pqr \\ 3 & r & pqr \end{vmatrix} = pqr\begin{vmatrix} 1 & p & 1 \\ 2 & q & 1 \\ 3 & r & 1 \end{vmatrix} \xrightarrow{R_3=R_3-3R_1} pqr\begin{vmatrix} 1 & p & 1 \\ 2 & q & 1 \\ 0 & r-3p & -2 \end{vmatrix}$$

$$\xrightarrow{R_2=R_2-2R_1} pqr\begin{vmatrix} 1 & p & 1 \\ 0 & q-2p & -1 \\ 0 & r-3p & -2 \end{vmatrix} = \frac{1}{2}pqr\begin{vmatrix} 1 & p & 1 \\ 0 & 2q-4p & -2 \\ 0 & r-3p & -2 \end{vmatrix}$$

$$\xrightarrow{R_2=R_2-R_3} \frac{1}{2}pqr\begin{vmatrix} 1 & p & 1 \\ 0 & 2q-p-r & 0 \\ 0 & r-3p & -2 \end{vmatrix}$$

$$= \frac{1}{2}pqr(1(2q-p-r)(-2)) = pqr(p-2q+r).$$

20 b)

$$\begin{vmatrix} 1 & p & q+r \\ 2 & q & p+r \\ 3 & r & p+q \end{vmatrix} \xrightarrow{R_3=R_3+R_2} \begin{vmatrix} 1 & p & p+q+r \\ 2 & q & p+q+r \\ 3 & r & p+q+r \end{vmatrix}$$

$$= (p+q+r)\begin{vmatrix} 1 & p & 1 \\ 2 & q & 1 \\ 3 & r & 1 \end{vmatrix} \xrightarrow{R_2=R_2-2R_1} (p+q+r)\begin{vmatrix} 1 & p & 1 \\ 0 & q-2p & -1 \\ 3 & r & 1 \end{vmatrix}$$

$$\xrightarrow{R_3=R_3-3R_1} (p+q+r)\begin{vmatrix} 1 & p & 1 \\ 0 & q-2p & -1 \\ 0 & r-3p & -2 \end{vmatrix}$$

$$= (p+q+r)((q-2p)(-2) - (-1)(r-3p))$$

$$= (p+q+r)(-2q+4p+r-3p) = (p+q+r)(p-2q+r).$$

20 c)

$$\begin{vmatrix} 1 & p & pr \\ 2 & q & qr \\ 3 & r & pq \end{vmatrix} \xrightarrow{R_2=R_2-2R_1} \begin{vmatrix} 1 & p & pr \\ 0 & q-2p & qr-2pr \\ 3 & r & pq \end{vmatrix} \xrightarrow{R_3=R_3-3R_1} \begin{vmatrix} 1 & p & pr \\ 0 & q-2p & r(q-2p) \\ 0 & r-3p & pq-3pr \end{vmatrix}$$

$$= (q-2p)\begin{vmatrix} 1 & p & pr \\ 0 & 1 & r \\ 0 & r-3p & pq-3pr \end{vmatrix}$$
$$= (q-2p)((pq-3pr)-r(r-3p))$$
$$= (q-2p)(pq-3pr-r^2+3pr) = (q-2p)(pq-r^2).$$

21 a)

$$\begin{vmatrix} p-q-r & p-q-r & p-q-r \\ p-q-r & 2p-q-r & p+3q-r \\ p-q-r & p-q & p-q \end{vmatrix} = (p-q-r)\begin{vmatrix} 1 & p-q-r & p-q-r \\ 1 & 2p-q-r & p+3q-r \\ 1 & p-q & p-q \end{vmatrix}$$
$$\xrightarrow{C_3=C_3-C_2} (p-q-r)\begin{vmatrix} 1 & p-q-r & 0 \\ 1 & 2p-q-r & -p+4q-r \\ 1 & p-q & 0 \end{vmatrix}$$
$$= (p-q-r)(-p+4q-r)\begin{vmatrix} 1 & p-q-r & 0 \\ 1 & 2p-q-r & 1 \\ 1 & p-q & 0 \end{vmatrix}$$
$$\xrightarrow{C_1\leftrightarrow C_3} -(p-q-r)(-p+4q-r)\begin{vmatrix} 0 & p-q-r & 1 \\ -p+4q-r & 2p-q-r & 1 \\ 0 & p-q & 1 \end{vmatrix}$$
$$\xrightarrow{R_1\leftrightarrow R_2} (p-q-r)(-p+4q-r)\begin{vmatrix} 1 & 2p-q-r & 1 \\ 0 & p-q-r & 1 \\ 0 & p-q & 1 \end{vmatrix}$$
$$= (p-q-r)(-p+4q-r)((p-q-r)\cdot 1 - 1(p-q))$$
$$= (p-q-r)(-p+4q-r)(p-q-r-p+q)$$
$$= -r(p-q-r)(-p+4q-r) = r(p-q-r)(p-4q+r).$$

21 b)

$$\begin{vmatrix} -5q-7r & -3p-3q+3r & 2p-2q-4r \\ 3p-q-3r & p+q-r & 2p-2q-4r \\ p-6q-9r & -3p-3q+3r & 3p-3q-6r \end{vmatrix} = \begin{vmatrix} -5q-7r & -3(p+q-r) & 2(p-q-2r) \\ 3p-q-3r & p+q-r & 2(p-q-2r) \\ p-6q-9r & -3(p+q-r) & 3(p-q-2r) \end{vmatrix}$$
$$= (p+q-r)\begin{vmatrix} -5q-7r & -3 & 2(p-q-2r) \\ 3p-q-3r & 1 & 2(p-q-2r) \\ p-6q-9r & -3 & 3(p-q-2r) \end{vmatrix}$$
$$= (p+q-r)(p-q-2r)\begin{vmatrix} -5q-7r & -3 & 2 \\ 3p-q-3r & 1 & 2 \\ p-6q-9r & -3 & 3 \end{vmatrix}$$
$$\xrightarrow{\text{rotate columns one step right}} (p+q-r)(p-q-2r)\begin{vmatrix} -3 & 2 & -5q-7r \\ 1 & 2 & 3p-q-3r \\ -3 & 3 & p-6q-9r \end{vmatrix}$$
$$= -3(p+q-r)(p-q-2r)\begin{vmatrix} 1 & -\frac{2}{3} & \frac{5}{3}q+\frac{7}{3}r \\ 1 & 2 & 3p-q-3r \\ -3 & 3 & p-6q-9r \end{vmatrix}$$

$$\xrightarrow{R_2=R_2-R_1} -3(p+q-r)(p-q-2r)\begin{vmatrix} 1 & -\frac{2}{3} & \frac{5}{3}q+\frac{7}{3}r \\ 0 & \frac{8}{3} & 3p-\frac{8}{3}q-\frac{16}{3}r \\ -3 & 3 & p-6q-9r \end{vmatrix}$$

$$\xrightarrow{R_3=R_3+3R_1} -3(p+q-r)(p-q-2r)\begin{vmatrix} 1 & -\frac{2}{3} & \frac{5}{3}q+\frac{7}{3}r \\ 0 & \frac{8}{3} & 3p-\frac{8}{3}q-\frac{16}{3}r \\ 0 & 1 & p-q-2r \end{vmatrix}$$

$$= -8(p+q-r)(p-q-2r)\begin{vmatrix} 1 & -\frac{2}{3} & \frac{5}{3}q+\frac{7}{3}r \\ 0 & 1 & \frac{9}{8}p-q-2r \\ 0 & 1 & p-q-2r \end{vmatrix}$$

$$\xrightarrow{R_3=R_3-R_2} -8(p+q-r)(p-q-2r)\begin{vmatrix} 1 & -\frac{2}{3} & \frac{5}{3}q+\frac{7}{3}r \\ 0 & 1 & \frac{9}{8}p-q-2r \\ 0 & 0 & -\frac{1}{8}p \end{vmatrix}$$

$$= -8(p+q-r)(p-q-2r)\left(-\frac{1}{8}p\right) = p(p+q-r)(p-q-2r).$$

21 c)

$$\begin{vmatrix} 2p+q+r & p & p+r \\ -2p-2r & -p+q-r & -2p-2r \\ 3p+q+2r & 2p+r & 2p+2r \end{vmatrix} \xrightarrow{R_3=R_3-R_1} \begin{vmatrix} 2p+q+r & p & p+r \\ -2p-2r & -p+q-r & -2p-2r \\ p+r & p+r & p+r \end{vmatrix}$$

$$= (p+r)\begin{vmatrix} 2p+q+r & p & p+r \\ -2p-2r & -p+q-r & -2p-2r \\ 1 & 1 & 1 \end{vmatrix} \xrightarrow{C_1=C_1-C_2} (p+r)\begin{vmatrix} p+q+r & p & p+r \\ -p-q-r & -p+q-r & -2p-2r \\ 0 & 1 & 1 \end{vmatrix}$$

$$= (p+r)(p+q+r)\begin{vmatrix} 1 & p & p+r \\ -1 & -p+q-r & -2p-2r \\ 0 & 1 & 1 \end{vmatrix} \xrightarrow{R_2=R_2+R_1} (p+r)(p+q+r)\begin{vmatrix} 1 & p & p+r \\ 0 & q-r & -p-r \\ 0 & 1 & 1 \end{vmatrix}$$

$$\xrightarrow{C_2=C_2-C_3} (p+r)(p+q+r)\begin{vmatrix} 1 & -r & p+r \\ 0 & p+q & -p-r \\ 0 & 0 & 1 \end{vmatrix}$$

$$= (p+q)(p+r)(p+q+r).$$

21 d)

$$\begin{vmatrix} 2p-4q+r & 3p-3q & 3p-6q+3r \\ p-q & p-q & p-3q+2r \\ p-3q+r & 2p-2q & 2p-2q \end{vmatrix} = \begin{vmatrix} 2p-4q+r & 3(p-q) & 3p-6q+3r \\ p-q & p-q & p-3q+2r \\ p-3q+r & 2(p-q) & 2p-2q \end{vmatrix}$$

$$= (p-q)\begin{vmatrix} 2p-4q+r & 3 & 3p-6q+3r \\ p-q & 1 & p-3q+2r \\ p-3q+r & 2 & 2p-2q \end{vmatrix}$$

$$\xrightarrow{R_1=R_1-3R_2} (p-q)\begin{vmatrix} -p-q+r & 0 & 3q-3r \\ p-q & 1 & p-3q+2r \\ p-3q+r & 2 & 2p-2q \end{vmatrix}$$

$$\xrightarrow{R_3=R_3-2R_2} (p-q)\begin{vmatrix} -p-q+r & 0 & 3q-3r \\ p-q & 1 & p-3q+2r \\ -p-q+r & 0 & 4q-4r \end{vmatrix}$$

$$\xrightarrow{R_1\leftrightarrow R_2,\, C_1\leftrightarrow C_2} (p-q)\begin{vmatrix} 1 & p-q & p-3q+2r \\ 0 & -p-q+r & 3q-3r \\ 0 & -p-q+r & 4q-4r \end{vmatrix}$$

$$\xrightarrow{R_3=R_3-R_2} (p-q)\begin{vmatrix} 1 & p-q & p-3q+2r \\ 0 & -p-q+r & 3q-3r \\ 0 & 0 & q-r \end{vmatrix}$$

$$= (p-q)(-p-q+r)(q-r).$$

22 a)

$$\begin{vmatrix} p & p^2 & p^3 \\ q & q^2 & q^3 \\ r & r^2 & r^3 \end{vmatrix} = p\begin{vmatrix} 1 & p & p^2 \\ q & q^2 & q^3 \\ r & r^2 & r^3 \end{vmatrix} = pq\begin{vmatrix} 1 & p & p^2 \\ 1 & q & q^2 \\ r & r^2 & r^3 \end{vmatrix} = pqr\begin{vmatrix} 1 & p & p^2 \\ 1 & q & q^2 \\ 1 & r & r^2 \end{vmatrix}$$

$$\xrightarrow{R_2=R_2-R_1} pqr\begin{vmatrix}1 & p & p^2\\ 0 & q-p & q^2-p^2\\ 1 & r & r^2\end{vmatrix} \xrightarrow{R_3=R_3-R_1} pqr\begin{vmatrix}1 & p & p^2\\ 0 & q-p & (q-p)(q+p)\\ 0 & r-p & (r-p)(r+p)\end{vmatrix}$$

$$= (q-p)pqr\begin{vmatrix}1 & p & p^2\\ 0 & 1 & q+p\\ 0 & r-p & (r-p)(r+p)\end{vmatrix} = (q-p)(r-p)pqr\begin{vmatrix}1 & p & p^2\\ 0 & 1 & q+p\\ 0 & 1 & r+p\end{vmatrix}$$

$$\xrightarrow{R_3=R_3-R_2} (q-p)(r-p)pqr\begin{vmatrix}1 & p & p^2\\ 0 & 1 & q+p\\ 0 & 0 & r-q\end{vmatrix} = pqr(q-p)(r-p)(r-q).$$

22 b)

$$\begin{vmatrix}1 & p & p^3\\ 1 & q & q^3\\ 1 & r & r^3\end{vmatrix} \xrightarrow{R_2=R_2-R_1} \begin{vmatrix}1 & p & p^3\\ 0 & q-p & q^3-p^3\\ 1 & r & r^3\end{vmatrix} \xrightarrow{R_3=R_3-R_1} \begin{vmatrix}1 & p & p^3\\ 0 & q-p & (q-p)(q^2+pq+p^2)\\ 0 & r-p & (r-p)(r^2+pr+p^2)\end{vmatrix}$$

$$= (q-p)\begin{vmatrix}1 & p & p^3\\ 0 & 1 & q^2+pq+p^2\\ 0 & r-p & (r-p)(r^2+pr+p^2)\end{vmatrix}$$

$$= (q-p)(r-p)\begin{vmatrix}1 & p & p^3\\ 0 & 1 & q^2+pq+p^2\\ 0 & 1 & r^2+pr+p^2\end{vmatrix} \xrightarrow{R_3=R_3-R_2} (q-p)(r-p)\begin{vmatrix}1 & p & p^3\\ 0 & 1 & q^2+pq+p^2\\ 0 & 0 & pr-pq+r^2-q^2\end{vmatrix}$$

$$= (q-p)(r-p)\begin{vmatrix}1 & p & p^3\\ 0 & 1 & q^2+pq+p^2\\ 0 & 0 & p(r-q)+(r-q)(r+q)\end{vmatrix}$$

$$= (q-p)(r-p)(r-q)(p+q+r).$$

22 c)

$$\begin{vmatrix}1 & p^2 & p^3\\ 1 & q^2 & q^3\\ 1 & r^2 & r^3\end{vmatrix} \xrightarrow{R_2=R_2-R_1} \begin{vmatrix}1 & p^2 & p^3\\ 0 & q^2-p^2 & q^3-p^3\\ 1 & r^2 & r^3\end{vmatrix} \xrightarrow{R_3=R_3-R_1} \begin{vmatrix}1 & p^2 & p^3\\ 0 & (q-p)(q+p) & (q-p)(q^2+pq+p^2)\\ 0 & (r-p)(r+p) & (r-p)(r^2+pr+p^2)\end{vmatrix}$$

$$= (q-p)\begin{vmatrix}1 & p^2 & p^3\\ 0 & q+p & q^2+pq+p^2\\ 0 & (r-p)(r+p) & (r-p)(r^2+pr+p^2)\end{vmatrix}$$

$$= (q-p)(r-p)\begin{vmatrix}1 & p^2 & p^3\\ 0 & q+p & q^2+pq+p^2\\ 0 & r+p & r^2+pr+p^2\end{vmatrix}$$

$$\xrightarrow{R_3=R_3-R_2} (q-p)(r-p)\begin{vmatrix}1 & p^2 & p^3\\ 0 & q+p & q^2+pq+p^2\\ 0 & r-q & pr-pq+r^2-q^2\end{vmatrix}$$

$$= (q-p)(r-p)\begin{vmatrix}1 & p^2 & p^3\\ 0 & q+p & q^2+pq+p^2\\ 0 & r-q & p(r-q)+(r-q)(r+q)\end{vmatrix}$$

$$= (q-p)(r-p)(r-q)\begin{vmatrix}1 & p^2 & p^3\\ 0 & q+p & q^2+pq+p^2\\ 0 & 1 & p+q+r\end{vmatrix}$$

$$= (q-p)(r-p)(r-q)((q+p)(p+q+r) - (q^2+pq+p^2))$$
$$= (q-p)(r-p)(r-q)(pq+pr+qr).$$

22 d)

$$\begin{vmatrix} 1 & p & p^2(p+1) \\ 1 & q & q^2(q+1) \\ 1 & r & r^2(r+1) \end{vmatrix} = \begin{vmatrix} 1 & p & p^3+p^2 \\ 1 & q & q^3+q^2 \\ 1 & r & r^3+r^2 \end{vmatrix} \xrightarrow{R_2=R_2-R_1} \begin{vmatrix} 1 & p & p^3+p^2 \\ 0 & q-p & q^3-p^3+q^2-p^2 \\ 1 & r & r^3+r^2 \end{vmatrix}$$

$$\xrightarrow{R_3=R_3-R_1} \begin{vmatrix} 1 & p & p^3+p^2 \\ 0 & q-p & (q-p)(q^2+pq+p^2)+(q-p)(q+p) \\ 0 & r-p & (r-p)(r^2+pr+p^2)+(r-p)(r+p) \end{vmatrix}$$

$$= (q-p) \begin{vmatrix} 1 & p & p^3+p^2 \\ 0 & 1 & q^2+pq+p^2+q+p \\ 0 & r-p & (r-p)(r^2+pr+p^2)+(r-p)(r+p) \end{vmatrix}$$

$$= (q-p)(r-p) \begin{vmatrix} 1 & p & p^3+p^2 \\ 0 & 1 & q^2+pq+p^2+q+p \\ 0 & 1 & r^2+pr+p^2+r+p \end{vmatrix}$$

$$\xrightarrow{R_3=R_3-R_2} (q-p)(r-p) \begin{vmatrix} 1 & p & p^3+p^2 \\ 0 & 1 & q^2+pq+p^2+q+p \\ 0 & 0 & r^2-q^2+pr-pq+r-q \end{vmatrix}$$

$$= (q-p)(r-p) \begin{vmatrix} 1 & p & p^3+p^2 \\ 0 & 1 & q^2+pq+p^2+q+p \\ 0 & 0 & p(r-q)+(r-q)(r+q)+(r-q) \end{vmatrix}$$

$$= (q-p)(r-p)(r-q) \begin{vmatrix} 1 & p & p^3+p^2 \\ 0 & 1 & q^2+pq+p^2+q+p \\ 0 & 0 & p+q+r+1 \end{vmatrix}$$

$$= (q-p)(r-p)(r-q)(p+q+r+1).$$

23 a)

$$\begin{vmatrix} 1 & p & q+r \\ 2 & q & p+r \\ 3 & r & p+q \end{vmatrix} \xrightarrow{C_3=C_3+C_2} \begin{vmatrix} 1 & p & p+q+r \\ 2 & q & p+q+r \\ 3 & r & p+q+r \end{vmatrix}$$

$$= (p+q+r) \begin{vmatrix} 1 & p & 1 \\ 2 & q & 1 \\ 3 & r & 1 \end{vmatrix} \xrightarrow{R_2=R_2-2R_1} (p+q+r) \begin{vmatrix} 1 & p & 1 \\ 0 & q-2p & -1 \\ 3 & r & 1 \end{vmatrix}$$

$$\xrightarrow{R_3=R_3-3R_1} (p+q+r) \begin{vmatrix} 1 & p & 1 \\ 0 & q-2p & -1 \\ 0 & r-3p & -2 \end{vmatrix} \xrightarrow{R_2=R_2-\frac{1}{2}R_3} (p+q+r) \begin{vmatrix} 1 & p & 1 \\ 0 & q-\frac{1}{2}p-\frac{1}{2}r & 0 \\ 0 & r-3p & -2 \end{vmatrix}$$

$$= (p+q+r)(p-2q+r).$$

23 b)

$$\begin{vmatrix} 1 & p+q & r \\ 2 & q+r & p \\ 4 & p+r & q \end{vmatrix} \xrightarrow{C_2=C_2+C_3} \begin{vmatrix} 1 & p+q+r & r \\ 2 & p+q+r & p \\ 4 & p+q+r & q \end{vmatrix}$$

$$= (p+q+r)\begin{vmatrix}1 & 1 & r\\ 2 & 1 & p\\ 4 & 1 & q\end{vmatrix} \xrightarrow{R_2=R_2-2R_1} (p+q+r)\begin{vmatrix}1 & 1 & r\\ 0 & -1 & p-2r\\ 4 & 1 & q\end{vmatrix}$$

$$\xrightarrow{R_3=R_3-4R_1} (p+q+r)\begin{vmatrix}1 & 1 & r\\ 0 & -1 & p-2r\\ 0 & -3 & q-4r\end{vmatrix} \xrightarrow{R_3=R_3-3R_2} (p+q+r)\begin{vmatrix}1 & 1 & r\\ 0 & -1 & p-2r\\ 0 & 0 & q+2r-3p\end{vmatrix}$$

$$= (p+q+r)(3p-q-2r).$$

23 c)

$$\begin{vmatrix}1 & 3 & 6\\ p+2r & q+2p & r+2q\\ q-r & r-p & p-q\end{vmatrix} \xrightarrow{R_2=R_2+R_3} \begin{vmatrix}1 & 3 & 6\\ p+q+r & p+q+r & p+q+r\\ q-r & r-p & p-q\end{vmatrix}$$

$$= (p+q+r)\begin{vmatrix}1 & 3 & 6\\ 1 & 1 & 1\\ q-r & r-p & p-q\end{vmatrix} \xrightarrow{C_2=C_2-3C_1} (p+q+r)\begin{vmatrix}1 & 0 & 6\\ 1 & -2 & 1\\ q-r & 4r-p-3q & p-q\end{vmatrix}$$

$$\xrightarrow{C_3=C_3-6C_1} (p+q+r)\begin{vmatrix}1 & 0 & 0\\ 1 & -2 & -5\\ q-r & 4r-p-3q & p-7q+6r\end{vmatrix}$$

$$\xrightarrow{C_3=C_3-\frac{5}{2}C_2} (p+q+r)\begin{vmatrix}1 & 0 & 0\\ 1 & -2 & 0\\ q-r & 4r-p-3q & \frac{7}{2}p+\frac{1}{2}q-4r\end{vmatrix}$$

$$= (p+q+r)(8r-7p-q).$$

23 d)

$$\begin{vmatrix}p-2r & q-2p & r-2q\\ 2 & 4 & 8\\ q+3r & r+3p & p+3q\end{vmatrix} \xrightarrow{R_1=R_1+R_3} \begin{vmatrix}p+q+r & p+q+r & p+q+r\\ 2 & 4 & 8\\ q+3r & r+3p & p+3q\end{vmatrix}$$

$$= (p+q+r)\begin{vmatrix}1 & 1 & 1\\ 2 & 4 & 8\\ q+3r & r+3p & p+3q\end{vmatrix} \xrightarrow{C_2=C_2-2C_1} (p+q+r)\begin{vmatrix}1 & -1 & \\ 2 & 0 & \\ q+3r & 3p-2q-5r & p-\end{vmatrix}$$

$$\xrightarrow{C_3=C_3-4C_1} (p+q+r)\begin{vmatrix}1 & -1 & -3\\ 2 & 0 & 0\\ q+3r & 3p-2q-5r & p-q-12r\end{vmatrix} \xrightarrow{C_3=C_3-3C_2} (p+q+r)\begin{vmatrix}1 & \\ 2 & \\ q+ & \end{vmatrix}$$

$$= 2(p+q+r)(5q+3r-8p).$$

24)

$$xy\begin{vmatrix}1 & 2 & 5\\ 0 & 1 & -3\\ 0 & 0 & 1\end{vmatrix} \xrightarrow{C_2=xC_2} y\begin{vmatrix}1 & 2x & 5\\ 0 & x & -3\\ 0 & 0 & 1\end{vmatrix} \xrightarrow{C_3=yC_3} \begin{vmatrix}1 & 2x & 5y\\ 0 & x & -3y\\ 0 & 0 & y\end{vmatrix}$$

$$\xrightarrow{C_1=C_1+C_3} \begin{vmatrix}1+5y & 2x & 5y\\ -3y & x & -3y\\ y & 0 & y\end{vmatrix} \xrightarrow{R_3=R_3-R_2} \begin{vmatrix}1+5y & 2x & 5y\\ -3y & x & -3y\\ 4y & -x & 4y\end{vmatrix}$$ as required.

25)

$$xy(x-1)(y-1)\begin{vmatrix}1&0&1\\-1&1&-1\\0&-1&-1\end{vmatrix}=\begin{vmatrix}x(y-1)&0&y\\x(1-y)&x-1&-y\\0&-(x-1)&y\end{vmatrix}$$

$$\xrightarrow{C_2=C_2+C_3}\begin{vmatrix}xy-x&y&y\\x-xy&x-y-1&-y\\0&1-x+y&y\end{vmatrix}\xrightarrow{R_3=R_3+R_2}\begin{vmatrix}xy-x&y&y\\x-xy&x-y-1&-y\\x-xy&0&0\end{vmatrix}$$

$$\xrightarrow{C_3=C_3+C_1}\begin{vmatrix}xy-x&y&xy-x+y\\x-xy&x-y-1&x-xy-y\\x-xy&0&x-xy\end{vmatrix}.$$

26)

$$|A|=0\Rightarrow\begin{vmatrix}6&3x+3a+5&-5\\5&4x+4a+4&-4\\4a-x+1&3a-2x+1&x-4a-1\end{vmatrix}=0$$

$$\xrightarrow{C_1=C_1+C_3}\begin{vmatrix}1&3x+3a+5&-5\\1&4x+4a+4&-4\\0&3a-2x+1&x-4a-1\end{vmatrix}\xrightarrow{R_2=R_2-R_1}\begin{vmatrix}1&3x+3a+5&-5\\0&x+a-1&1\\0&3a-2x+1&x-4a-1\end{vmatrix}$$

$$\xrightarrow{C_2=C_2+C_3}\begin{vmatrix}1&3(x+a)&-5\\0&x+a&1\\0&-(x+a)&x-4a-1\end{vmatrix}=(x+a)\begin{vmatrix}1&3&-5\\0&1&1\\0&-1&x-4a-1\end{vmatrix}$$

$$\xrightarrow{C_3=C_3-C_2}(x+a)\begin{vmatrix}1&3&-8\\0&1&0\\0&-1&x-4a\end{vmatrix}=(x+a)(x-4a)=0,\quad x=-a\text{ or }4a.$$

27)

$$|A|=0\Rightarrow\begin{vmatrix}-2a&6x+24a&-x-a\\2x-4a&7x+28a&3x-5a\\3x-7a&13x+52a&4x-8a\end{vmatrix}=0\xrightarrow{C_1=C_1-C_3}\begin{vmatrix}x-a&6x+24a&-x-a\\-x+a&7x+28a&3x-5a\\-x+a&13x+52a&4x-8a\end{vmatrix}$$

$$=(x-a)\begin{vmatrix}1&6x+24a&-x-a\\-1&7x+28a&3x-5a\\-1&13x+52a&4x-8a\end{vmatrix}\xrightarrow{R_2=R_2+R_1,\ R_3=R_3+R_1}(x-a)\begin{vmatrix}1&6x+24a&-x-a\\0&13x+52a&2x-6a\\0&19x+76a&3x-9a\end{vmatrix}$$

$$\xrightarrow{R_3=R_3-R_2}(x-a)\begin{vmatrix}1&6x+24a&-x-a\\0&13x+52a&2x-6a\\0&6x+24a&x-3a\end{vmatrix}\xrightarrow{R_2=R_2-2R_3}(x-a)\begin{vmatrix}1&6x+24a&-x-a\\0&x+4a&0\\0&6x+24a&x-3a\end{vmatrix}$$

$$=(x-a)(x+4a)(x-3a)=0,\quad x=a,-4a,3a.$$

28 a)

$$\begin{vmatrix}0&1&x-p\\x&p&px\\p&q&p^2\end{vmatrix}\xrightarrow{C_3=C_3-pC_1}\begin{vmatrix}0&1&x-p\\x&p&0\\p&q&0\end{vmatrix}.$$

28 b)

$$|A| = 0 \Rightarrow \begin{vmatrix} 0 & 1 & x-p \\ x & p & 0 \\ p & q & 0 \end{vmatrix} = 0 \Rightarrow (x-p)(qx-p^2) = 0 \Rightarrow x = p \text{ or } x = \frac{p^2}{q}.$$

29)

$$\begin{vmatrix} x^3 & 0 & p^2x \\ q & pq & q \\ x+q & p^2+pq & p+q \end{vmatrix} \xrightarrow{C_1=C_1-C_3} \begin{vmatrix} x^3-p^2x & 0 & p^2x \\ 0 & pq & q \\ x-p & p^2+pq & p+q \end{vmatrix}$$

$$= \begin{vmatrix} x(x-p)(x+p) & 0 & p^2x \\ 0 & pq & q \\ x-p & p^2+pq & p+q \end{vmatrix} = (x-p) \begin{vmatrix} x(x+p) & 0 & p^2x \\ 0 & pq & q \\ 1 & p^2+pq & p+q \end{vmatrix}$$

$$\xrightarrow{C_3=C_3-\frac{1}{p}C_2} (x-p) \begin{vmatrix} x(x+p) & 0 & p^2x \\ 0 & pq & 0 \\ 1 & p^2+pq & 0 \end{vmatrix} = p^2x(x-p) \begin{vmatrix} x(x+p) & 0 & 1 \\ 0 & pq & 0 \\ 1 & p^2+pq & 0 \end{vmatrix}$$

$$= p^2x(x-p)(-pq) = p^3qx(p-x).$$

30)

$$\begin{vmatrix} q-x & -p^2 & x-q \\ x-q & px & q-p \\ x-q & p^2 & p+2q-x \end{vmatrix} \xrightarrow{R_3=R_3+R_1} \begin{vmatrix} q-x & -p^2 & x-q \\ x-q & px & q-p \\ 0 & 0 & p+q \end{vmatrix}$$

$$= \begin{vmatrix} -(x-q) & -p^2 & x-q \\ x-q & px & q-p \\ 0 & 0 & p+q \end{vmatrix}$$

$$= (x-q) \begin{vmatrix} -1 & -p^2 & x-q \\ 1 & px & q-p \\ 0 & 0 & p+q \end{vmatrix} \xrightarrow{R_2=R_2+R_1} (x-q) \begin{vmatrix} -1 & -p^2 & x-q \\ 0 & p(x-p) & q-x \\ 0 & 0 & p+q \end{vmatrix}$$

$$= (x-q)(p+q)(-p(x-p)) = p(p+q)(p-x)(x-q).$$

31 a)

$$\begin{vmatrix} a-2x+b & x+a & 4a-2x+b \\ a-2x+2b-1 & x+a+1 & 4a-2x+3b \\ 3a+b-x-1 & 1 & 6a+2b-2x \end{vmatrix} \xrightarrow{R_1=R_1-R_2} \begin{vmatrix} 1-b & -1 & -2b \\ a-2x+2b-1 & x+a+1 & 4a-2x+3b \\ 3a+b-x-1 & 1 & 6a+2b-2x \end{vmatrix}$$

$$\xrightarrow{C_1=C_1+C_2} \begin{vmatrix} -b & -1 & -2b \\ 2a-x+2b & x+a+1 & 4a-2x+3b \\ 3a-x & 1 & 6a+2b-2x \end{vmatrix} \xrightarrow{R_3=R_3+R_1} \begin{vmatrix} - \\ 2a-x \\ 3a \end{vmatrix}$$

$$= (3a-x) \begin{vmatrix} -b & -1 & -2b \\ 2a-x+2b & x+a+1 & 4a-2x+3b \\ 1 & 0 & 2 \end{vmatrix} \xrightarrow{C_3=C_3-2C_1} (3a-$$

$$= -b(3a-x)\begin{vmatrix} -b & -1 & 0 \\ 2a-x+2b & x+a+1 & 1 \\ 1 & 0 & 0 \end{vmatrix} \xrightarrow{\text{rotate rows down}} b(3a-x).$$

31 b)

$$a = 6, \quad 6 + b = 13 \Rightarrow b = 7, \quad |A| = \pm\frac{1260}{300} = \pm\frac{21}{5}.$$

$$|A| = \frac{21}{5}: \quad 7(18-x) = \frac{21}{5} \Rightarrow 18 - x = \frac{3}{5} \Rightarrow x = \frac{87}{5}.$$

$$|A| = -\frac{21}{5}: \quad 7(18-x) = -\frac{21}{5} \Rightarrow 18 - x = -\frac{3}{5} \Rightarrow x = \frac{93}{5}.$$

$$\therefore x = \frac{87}{5} \text{ or } \frac{93}{5}.$$